

Regional synapse gain and loss accompany memory formation in larval zebrafish

DOI: <https://doi.org/10.1073/pnas.2107661119>

Previous Literature

Learning and memory have been hypothesised to depend on structural changes at synapses for a long time. This idea was initially proposed by Ramón y Cajal who suggested that neuronal connections must change during memory formation (Cajal, 1894). In the late 20th century, Bliss and Lomo's discovery of long term potentiation (LTP) in hippocampal slices provided the first physiological evidence for long lasting synaptic strengthening (Bliss & Lomo, 1973). Subsequent slice studies characterised molecular mechanisms of LTP, implicating postsynaptic insertion of AMPA receptors and dendritic spine enlargement as correlates of potentiation (Malinow & Malenka, 2002). However, these ex vivo studies do not capture synaptic remodeling in circuits during actual learning within the organism.

Further progress was made with the introduction of in vivo two-photon imaging in rodents. Holtmaat et al. (2005) showed that novel sensory experience increased formation of new dendritic spines and loss of old ones in the somatosensory cortex of mice. Yang et al. (2009) found that motor learning induces rapid spine formation in the motor cortex, with a subset of new spines persisting for weeks. This suggests that spine dynamics correlate with memory retention. Meanwhile, Hayashi-Takagi et al. (2015) used optogenetic tagging to identify single synapses potentiated by fear conditioning in the amygdala, revealing that increased spine size at those synapses underlay conditioned fear memories. Yet, most of these studies focused on cortical/ hippocampal regions or focus on single synapse changes. Whereas, direct, longitudinal

imaging of synapses in deep structures such as the amygdala during associative learning remained technically challenging and therefore mostly unexplored.

Classical/ Pavlovian conditioning experiments in rodents have linked synaptic changes in the basolateral amygdala to auditory fear conditioning (Rogan et al., 1997). Electrophysiological recordings have demonstrated increased AMPA/NMDA current ratios in amygdalar neurons after conditioning (McKernan & Shinnick-Gallagher, 1997), implying postsynaptic strengthening. Contextual fear conditioning engaged hippocampal engram neurons, which exhibited larger and more densely packed spines than non-engram cells (Liu et al., 2012). These studies, however, relied on endpoint analyses rather than imaging the same synapses over time, and did not provide unbiased, whole-region synapse maps.

Overall, this field had amassed strong evidence that synaptic remodeling occurs with learning in ex vivo slices and in vivo superficial regions, but lacked unbiased, longitudinal, micro level mapping of structural changes in situ within a deep, behaviorally relevant structure during associative memory formation. This gap motivated Dempsey et al. (2022) to develop tools for capturing excitatory synapse dynamics in larval zebrafish pallium before and after classical conditioning in order to link structural plasticity in the synaptic level to learning in a living vertebrate brain. Zebrafish larvae offer an attractive model for high resolution imaging in an intact vertebrate. Their pallium is a homologous structure to our amygdala, larvae are transparent, and small enough for selective plane illumination microscopy (SPIM) (Ahrens et al., 2013). Prior work established associative learning in adult zebrafish and showed place preference learning in larvae, but synaptic changes accompanying learning in larval zebrafish remained unexplored (Valente et al., 2012).

Study Objectives

Dempsey et al. aimed to determine how excitatory synapse distributions change during associative memory formation within the zebrafish pallium. The main objectives of the study included:

- ❖ To image the same synapses longitudinally (over time) in a living vertebrate across learning.
- ❖ To provide whole-region, unbiased maps of synapse gain and loss.
- ❖ To link these structural changes to behaviorally validated memory formation in a deep brain structure.

Model System and Experimental Paradigm

The researchers used young zebrafish larvae that were 14 to 16 days post-fertilization (dpf). These fish were genetically modified to produce a special fluorescent marker called PSD-95.FingR-GFP. This marker attaches to PSD-95, a protein found at excitatory synapses and glows green allowing us to track individual synapses. This marker binds in a way that doesn't interfere with the functioning of the synapse, so brain activity remains natural (Gross et al., 2013). To carry out the learning experiment, the larvae were gently placed in agarose which was used to keep their heads still for imaging, while their tails were left free to move. The fish were trained using a method called tail-flick conditioning (TFC), which is a form of classical conditioning. TFC involved using a mild heat pulse (unconditioned stimulus) from an infrared laser which naturally caused the fish to flick their tail (response) to be paired with a harmless green light (conditioned stimulus) for conditioning.

The researchers measured the fish's tail flicks using automated tracking software, and based on their responses, grouped the fish into the following categories:

- ❖ Learners (L) – fish that learned to flick their tails in response to the light alone
- ❖ Partial learners (PL) – fish that showed some learning
- ❖ Non-learners (NL) – fish that didn't learn the association
- ❖ Control groups – fish that were only exposed to the light, only to the heat, or to nothing

To prove that the zebrafish truly formed a memory, the researchers tested whether NMDA receptors were required in this process. These NMDA receptors help control synaptic plasticity – the brain's ability to strengthen or weaken connections between neurons during learning. The researchers used a chemical called APV, which blocks NMDA receptors. When the fish were given APV, TFC did not work. They also observed that even after the fish learned the association, the response gradually faded if the light was shown repeatedly without the heat. This extinction is also a classic feature of classical conditioning. It shows that the fish were not just reacting instinctively or reflexively, but were actually forming and updating a memory based on experience.

To study how the synapses changed with learning, the scientists used a custom-built light-sheet microscope (SPIM) designed for imaging zebrafish. This microscope provided high-resolution 3D images (0.5 μm across and 1.3 μm deep) while also minimizing light damage to the brain. Each fish was imaged twice. Once before training, and again about 5 hours after the learning task. The images focused on the left pallium region of the zebrafish brain. The glowing PSD-95 spots (puncta) – representing individual excitatory synapses – were detected using computer algorithms that identified the bright points. These results were also checked by trained human observers who didn't know the experimental conditions (to avoid bias). On average, the

researchers recorded about 1,155 synapses per fish, which is about 0.5% of all the synapses in the pallium. Which gives us a good idea of how synapse patterns changed during learning.

Results & Interpretation

- ❖ Across L, PL, NL, and controls, the total number of PSD-95.FingR labeled synapses in the pallium did not change significantly post TFC, indicating that overall synapse recycling (gain and loss) balanced out in learners.
- ❖ By aligning synapse maps to a common template and pairing pre-/post-TFC puncta within 4 μm , the authors identified “lost” and “gained” synapses. Support vector machine–derived decision boundary (DB) planes revealed that in L fish, synapse gains were concentrated ventrolaterally, while losses predominated dorsomedially. No such spatial segregation occurred in NL or controls; PL fish showed a milder pattern.
- ❖ Median PSD-95.FingR–GFP intensity per punctum (which was used as a proxy for synaptic strength) did not change significantly with learning, either overall or regionally. Simulations showed that intensity fluctuations alone could not account for detected gain/loss patterns.

Limitations & Conclusion

Though the study provides us with strong findings, several limitations should be considered when interpreting them. First, the imaging captured only ~0.5% of all synapses in the zebrafish pallium, and the fluorescent labeling was mosaic, this meant that only a subset of neurons expressed the PSD-95.FingR-GFP marker. This introduces the possibility that the sampled synapses do not necessarily reflect the exact distribution or diversity of the synaptic changes throughout the brain. Such sampling bias could lead to over or underestimation of

region specific gain/loss patterns. Second, the detection of changes in synaptic strength was compromised by the sensitivity and resolution of the PSD-95.FingR-GFP probe. Although simulations confirmed that intensity fluctuations could not explain the spatial patterns of synapse gain and loss, small but functionally meaningful changes in PSD-95 levels could fall below detection thresholds. Moreover, the study only examined excitatory synapses marked by PSD-95; inhibitory synapses and presynaptic components were not visualized, leaving out potentially important parts of learning related plasticity (El-Husseini et al., 2002). Third, the imaging of each fish was only done at two time intervals, prior to and around five hours post-conditioning. This comparatively low temporal resolution allows transient changes in the synaptic structures, or delayed remodeling during memory consolidation to be overlooked. Future research can incorporate better temporal sampling, presynaptic markers, and genetic engram tagging to further elucidate the cellular and molecular mechanisms relating structural plasticity to memory storage.

Overall, this study provides compelling evidence that associative memory formation in larval zebrafish is accompanied by spatially distinct structural changes in the brain. Through high resolution imaging, the research demonstrates to us that learning does not change the total number of synapses, but instead results in regional synapse gain in the ventrolateral pallium and loss in the dorsomedial pallium, a region functionally homologous to the mammalian amygdala. These findings highlight how memory formation involves selective remodeling of neural circuits, providing insights into the physical basis of learning. This research opens opportunities for future studies on memory mechanisms in transparent vertebrate models.

References

- Ahrens, M. B., Orger, M. B., Robson, D. N., Li, J. M., & Keller, P. J. (2013). Whole-brain functional imaging at cellular resolution using light-sheet microscopy. *Nature methods*, 10(5), 413–420. <https://doi.org/10.1038/nmeth.2434>
- Bliss, T. V., & Lømo, T. (1973). Long-lasting potentiation of synaptic transmission in the dentate area of the anaesthetized rabbit following stimulation of the perforant path. *The Journal of Physiology*, 232(2), 331–356. <https://doi.org/10.1113/jphysiol.1973.sp010273>
- Cajal, S. R. y. (1894). *Histology of the nervous system of man and vertebrates*. Oxford University Press.
- Dempsey, Z. Du, A. Nadtochiy, C.D. Smith, K. Czajkowski, A. Andreev, D.N. Robson, J.M. Li, S. Applebaum, T.V. Truong, C. Kesselman, S.E. Fraser, & D.B. Arnold, Regional synapse gain and loss accompany memory formation in larval zebrafish, *Proc. Natl. Acad. Sci. U.S.A.* 119 (3) e2107661119, <https://doi.org/10.1073/pnas.2107661119> (2022)
- El-Husseini, A.el-D., Schnell, E., Dakoji, S., Sweeney, N., Zhou, Q., Prange, O., Gauthier-Campbell, C., Aguilera-Moreno, A., Nicoll, R. A., & Brecht, D. S. (2002). Synaptic strength regulated by palmitate cycling on PSD-95. *Cell*, 108(6), 849–863. [https://doi.org/10.1016/s0092-8674\(02\)00683-9](https://doi.org/10.1016/s0092-8674(02)00683-9)
- Gross, G. G., Junge, J. A., Mora, R. J., Kwon, H. B., Olson, C. A., Takahashi, T. T., Liman, E. R., Ellis-Davies, G. C. R., McGee, A. W., Sabatini, B. L., Roberts, R. W., & Arnold, D. B. (2013). Recombinant probes for visualizing endogenous synaptic proteins in living neurons. *Neuron*, 78(6), 971–985. <https://doi.org/10.1016/j.neuron.2013.04.017>

Holtmaat, A. J., Trachtenberg, J. T., Wilbrecht, L., Shepherd, G. M., Zhang, X., Knott, G. W., & Svoboda, K. (2005). Transient and persistent dendritic spines in the neocortex in vivo.

Neuron, 45(2), 279–291. <https://doi.org/10.1016/j.neuron.2005.01.003>

Hayashi-Takagi, A., Yagishita, S., Nakamura, M., Shirai, F., Wu, Y. I., Loshbaugh, A. L.,

Kuhlman, B., Hahn, K. M., & Kasai, H. (2015). Labelling and optical erasure of synaptic memory traces in the motor cortex. *Nature*, 525(7569), 333–338.

<https://doi.org/10.1038/nature15257>

Liu, X., Ramirez, S., Pang, P. T., Puryear, C. B., Govindarajan, A., Deisseroth, K., & Tonegawa,

S. (2012). Optogenetic stimulation of a hippocampal engram activates fear memory recall. *Nature*, 484(7394), 381–385. <https://doi.org/10.1038/nature11028>

Malinow, R., & Malenka, R. C. (2002). AMPA receptor trafficking and synaptic plasticity.

Annual Review of Neuroscience, 25, 103–126.

<https://doi.org/10.1146/annurev.neuro.25.112701.142758>

McKernan, M. G., & Shinnick-Gallagher, P. (1997). Fear conditioning induces a lasting potentiation of synaptic currents in vitro. *Nature*, 390(6660), 607–611.

<https://doi.org/10.1038/37605>

Rogan, M. T., Stäubli, U. V., & LeDoux, J. E. (1997). Fear conditioning induces associative long-term potentiation in the amygdala. *Nature*, 390(6660), 604–607.

<https://doi.org/10.1038/37601>

Valente, A., Huang, K. H., Portugues, R., & Engert, F. (2012). Ontogeny of classical and operant learning behaviors in zebrafish larvae. *Frontiers in Neural Circuits*, 6, 126.

<https://doi.org/10.1101/lm.025668.112>

Yang, G., Pan, F., & Gan, W. B. (2009). Stably maintained dendritic spines are associated with lifelong memories. *Nature*, 462(7275), 920–924. <https://doi.org/10.1038/nature08577>